

PROJECT: Employee Management System (EMS)

Tech Stack:

- Frontend: React + TypeScript + Vite + Tailwind CSS
- Backend: NestJS + TypeORM
- Database: PostgreSQL
- Authentication: JWT + bcrypt

CURRENT STATUS

DATABASE (PostgreSQL) Created and connected:

- users
- roles
- employees
- departments
- attendance
- leave_requests

Authentication data exists in users table.

BACKEND (NestJS)

Auth Module:

- POST /auth/register
- POST /auth/login
- JWT authentication implemented
- Password hashing using bcrypt
- Login returns access_token

AuthService:

register()

- hashes password
- saves user

login()

- validates email
- compares password with bcrypt
- returns JWT token

Users Module:

- findByEmail()
- create()

Departments Module:

- CRUD APIs implemented and tested

Employees Module:

- CRUD APIs implemented and tested

All APIs tested successfully in Postman.

Frontend (React)

Pages Created:

- Login.tsx
- AdminLogin.tsx
- EmployeeLogin.tsx
- AdminDashboard.tsx
- EmployeeDashboard.tsx

Routes:

/login /login/admin /login/employee /admin/dashboard /employee/dashboard

Authentication Integration

Created: src/api/auth.ts

Contains:

```
export async function login(email: string, password: string)
```

Admin Login:

- Connected to backend
- Calls POST `http://localhost:3000/auth/login`
- Stores `access_token` in `localStorage`
- Navigates to: `/admin/dashboard`

Employee Login:

- Connected to backend
- Calls same login API

- Stores access_token in localStorage
- Navigates to: /employee/dashboard

Current Working Flow

Admin Login: admin@ems.com admin123

Successfully logs in and opens:

- /admin/dashboard

Employee Login: Uses registered employee account

Successfully logs in and opens:

- /employee/dashboard

Issues Already Fixed

- Duplicate useState import
- Route mismatch errors
- No routes matched location errors
- Backend 401 issues
- Employee login state bindings
- Duplicate Sign In button
- Dashboard navigation
- LocalStorage token saving

Current Folder Structure

```
src/ |—— api/ | |—— auth.ts |—— pages/ | |—— Login.tsx | |—— AdminLogin.tsx | |——
EmployeeLogin.tsx | |—— AdminDashboard.tsx | |—— EmployeeDashboard.tsx |—— routes/ |
|—— AppRoutes.tsx
```

NEXT TASK

Continue Week 1 Frontend Work.

Priority: 1. Department CRUD Frontend

- Department List
- Add Department
- Edit Department
- Delete Department
- Connect with NestJS Department APIs
- Employee CRUD Frontend

- Employee List
- Add Employee
- Edit Employee
- Delete Employee
- Connect with NestJS Employee APIs

IMPORTANT

In the next chat:

- Ask me which file you want.
- I will send the complete file.
- Tell me exactly what code to replace.
- Work step by step.
- Do not rewrite entire project at once.
- Make one change, test it, then move to next step.